

COL362/632 Introduction to Database Management Systems

Plan Cost Estimation / Cardinality Estimation

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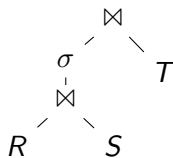
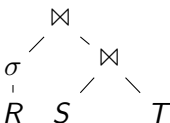
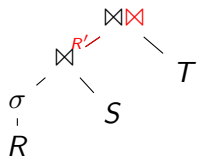


So far ...

- ▶ Declarative queries are parsed into expression trees
- ▶ Physical operator implementations for different operators
 - Algorithms
 - Cost analysis
- ▶ Lets now look at **cost estimation**

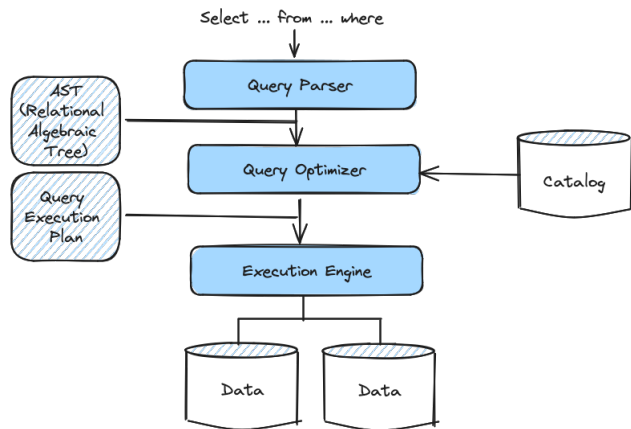
Cost Estimation Problem

- ▶ The goal of the query optimizer is to **find a physical plan that has least cost**
- ▶ For this, we need to **estimate** the total cost
 - Estimate cost of each operation in plan tree
 - We've already discussed these for various operators
 - Depends on input cardinalities
- ▶ Consider three (equivalent) plans //we'll discuss equivalence later



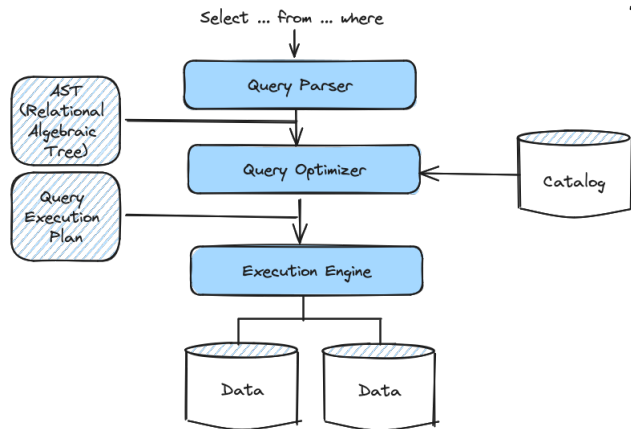
- ▶ We know how to compute the cost if we know the cardinalities
 - $\text{cost}(R' \bowtie T) = 3(b_{R'} + b_T)$ //e.g., hash join
 - $b_{R'} = \frac{n_{R'}}{b}$ // b is block size
 - $n_{R'} = n_{\sigma(R) \bowtie S}$
- ▶ Cardinality estimation problem: e.g., **estimate** $n_{\sigma(R) \bowtie S}$

Catalogue



- ▶ **Collect** statistical summaries of stored data
- ▶ **Estimate size** (cardinalities) in a bottom-up fashion
 - This is the most difficult part!
- ▶ **Estimate cost** using the estimated size
 - Formulas, heuristics (recall operator implementation)

Catalogue



Statistics stored in Catalogue

- ▶ n_R //no. of tuples in R
- ▶ b_R //no. of pages
- ▶ ℓ_R //size of $r \in R$
- ▶ $V(R, A)$ // no. of distinct values
- ▶ $\min(R, A), \max(R, A)$
- ▶ Indexes and metadata (e.g., height, keys)
- ▶ Histograms, Sketches,...

Statics are collected periodically, using sampling

Cardinality Estimation Problem

- ▶ Consider a SPJ query Q

$Q \equiv \text{select } \dots \text{ from } R_1, \dots, R_n \text{ where } \text{cond}_1 \text{ and } \dots \text{ cond}_k$

- ▶ Given $n_{R_1}, n_{R_2}, \dots, n_{R_n}$

- ▶ Estimate n_Q

- ▶ Note: this does not have to be exact! A **good approximate** is good enough

- ▶ Observe that $n_Q \leq n_{R_1} \times n_{R_2} \times \dots \times n_{R_n}$

- ▶ **Key idea:** Each condition reduces the size n_Q by **selectivity factor**

Example

- ▶ Consider the following query Q on relations $R(A, B)$, $S(B, C)$, and $T(C, D)$

and $n_R = 30K$, $n_S = 200K$, and $n_T = 10K$

- ▶ Let us assume that selectivity of

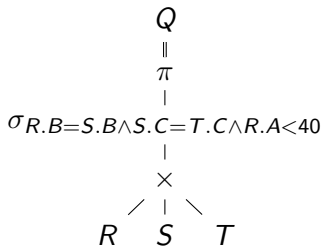
- $R.B = S.B$ is $1/3$
- $S.C = T.C$ is $1/10$
- $R.A < 40$ is $1/2$

- ▶ **Estimate** the size n_Q of the query Q

- ▶ $n_Q = 30K \times 200K \times 10K \times \frac{1}{3} \times \frac{1}{10} \times \frac{1}{2} = 10^{12}$

- ▶ **Note** This assumes that attributes are independent!

//more on this later



Selectivity Estimation

- ▶ $A = c$ $// \sigma_{A=c}(R)$
– selectivity = $\frac{1}{V(R,A)}$
- ▶ $A < c$ $// \sigma_{A < c}(R)$
– selectivity = $\frac{c - \min(R,A)}{\max(R,A) - \min(R,A)}$
- ▶ $A = B$ $// R \bowtie_{R.A=S.B} S$
– selectivity = $\frac{1}{\max(V(R,A), V(S,B))}$

Assumptions

- ▶ **Containment of values:** if $V(R, A) \leq V(S, B)$, then all values of $R.A$ occur in $S.B$
 - **Note:** this always holds when there is a PK-FK relationship
- ▶ **Preservation of values:** for any other attribute C , we have $V(R \bowtie_{A=B} S, C) = V(R, C) \text{ or } V(S, C)$
 - **Note:** We don't need this to estimate the size of the join, but we need it in estimating the next operator

Join Size Estimation

- ▶ Consider relations $R(A, \dots)$ and $S(B, \dots)$ and $Q \equiv R \bowtie_{R.A=S.B} S$
- ▶ Assume $V(R, A) \leq V(S, B)$
 - A tuple $r \in R$ joins with $n_S / V(S, B)$ tuple(s) in S
 - Hence, $n_Q = \frac{n_R * n_S}{V(S, B)}$
- ▶ size of join $n_Q = \frac{n_R * n_S}{\max(V(R, A), V(S, B))}$

Example

- ▶ $n_R = 10K$, $n_S = 20K$
- ▶ $V(R, A) = 100$, $V(S, A) = 200$
- ▶ Q : how large is $R \bowtie_{A=B} S$?

Computing Plan Cost

- ▶ Estimate cardinality in a bottom-up fashion
 - Cardinality is the size n_R of a relation R
 - Compute size of all intermediate relations in a plan
- ▶ Estimate cost by using the estimated cardinalities

Computing Plan cost (Example)

Supplier(sid, name, city, state)

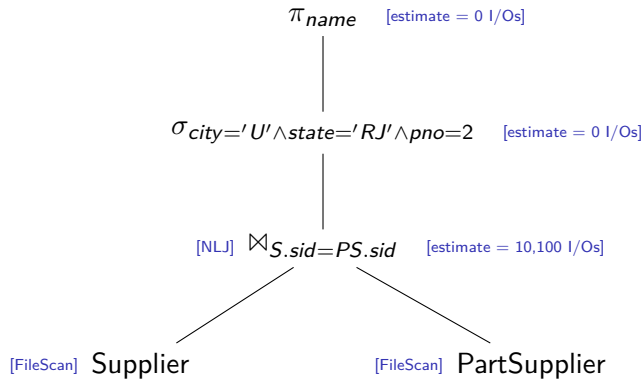
PartSupplier(sid,pno,quantity)

select name from Supplier S, PartSupplier PS where S.sid = PS.sid and PS.pno=2 and S.city = 'Udaipur' and S.state='RJ'

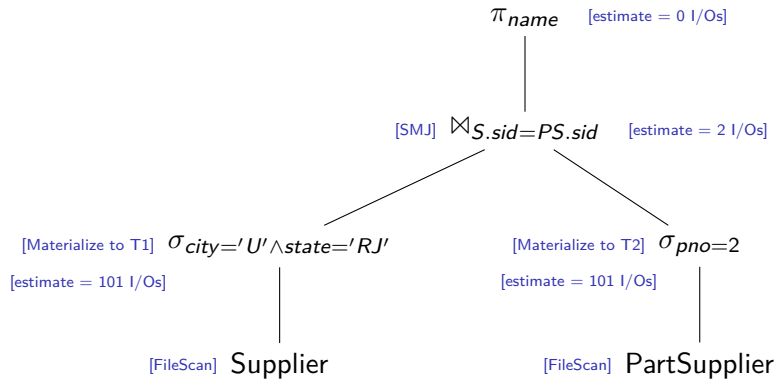
Catalogue

- ▶ $n_S = 1000$, $n_{PS} = 10,000$
- ▶ $b_S = 100$, $b_{PS} = 100$
- ▶ $V(S, city) = 20$, $V(S, state) = 10$, $V(PS, pno) = 2500$
- ▶ $B = 11$

Physical Query Plan 1



Physical Query Plan 2

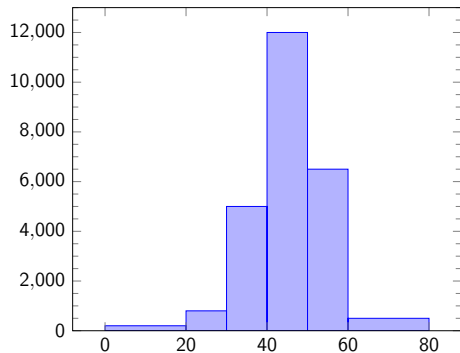


Histograms

A histogram is an approximate data synopsis that discretizes a value collection into different bins and stores the frequency of the values that fall into each bin

- ▶ Maintain an occurrence of count per value (or range of values)
- ▶ Makes cardinality estimation more accurate
(hence cost estimations are more accurate)

Histograms



age	0-19	20-29	30-39	40-49	50-59	60-80
#Tuples	200	800	5000	12000	6500	500

- ▶ `Employee(id, name, age)`
- ▶ $n_E = 25000$, $V(E, \text{age}) = 50$
 $\min(\text{age}) = 19$, $\max(\text{age}) = 68$
- ▶ **Estimate** size of $\sigma_{\text{age}=48}(E)$
– 1200
- ▶ **Estimate** size of $\sigma_{\text{age}>28 \wedge \text{age}<35}(E)$
– $1 \times 80 + 5 \times 500 = 2580$

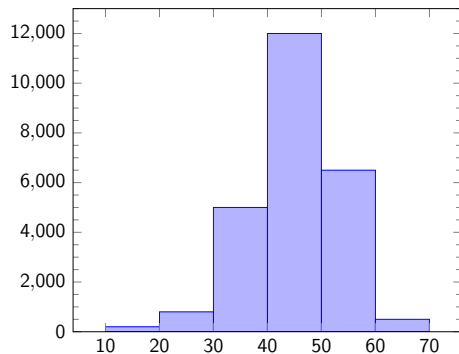
Histogram Types

Bucket boundaries have huge impact of estimations. How to determine bucket boundaries?

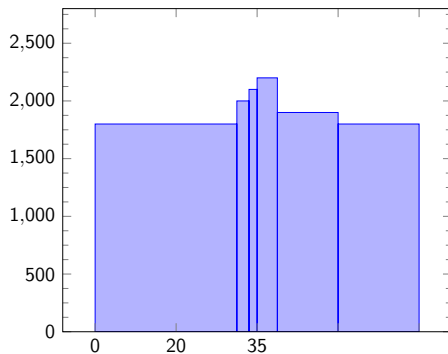
Types of histograms

- ▶ Equi-width
- ▶ Equi-depth
- ▶ Compressed
- ▶ V-Optimal

Equi-width vs Equi-depth Histograms



- ▶ all buckets have same width



- ▶ vary the width of buckets so that total number of tuples for each bucket is roughly the same
- ▶ “frequent zones” get their own bucket

V-optimal Histograms

- ▶ Define bucket boundaries in an optimal way, to minimize the error of overall point queries
- ▶ Computed rather expensively, using dynamic programming
- ▶ Modern database systems use V-optimal or some variations

Summary

- ▶ Compute selectivity for basic conditions
 - Formula-based
 - Histogram based (some databases also use sketches)
- ▶ Assumption – Uniform distribution
- ▶ Assumption – Independent conditions (attributes)
 - selectivity of AND = prod. of selectivity
 - selectivity of OR = sum of selectivity - prod. of selectivity
 - selectivity of NOT = 1 - selectivity

Quiz (Ungraded)

CarDekho Example

- ▶ Consider a relation cars (make, model, year, price)

Q Estimate the cardinality of

```
SELECT * FROM Cars WHERE make = 'Honda' AND model = 'Civic';
```

- ▶ Catalog

- $n_R = 100,000$
- Frequency of Honda is 10%
- Frequency of Civic is 5%

- ▶ If we assume independence, cardinality = $100,000 \times \frac{1}{10} \times \frac{1}{20} = 500$

- ▶ But this is wrong!

- ▶ In reality, **only Honda makes Civic**

- So, every row where Model = 'Civic' must also have Make = 'Honda'
- Correct estimated cardinality = $100,000 \times \frac{1}{20} = 5000$

Key Takeways

- ▶ Independency assumption fails because Make and Model are strongly correlated
- ▶ Assuming independence underestimates selectivities
- ▶ Advanced selectivity estimation
 - Multidimensional histograms
 - Bayesian Networks
 - Machine Learning
- ▶ This beyond COL362, check out Advance Data Management or Spl. Topics in Data(base) Systems